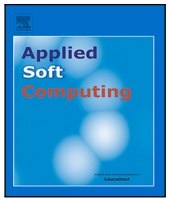
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A staged fuzzy evolutionary algorithm for constrained large-scale multiobjective optimization

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ABSTRACT

Constrained multiobjective optimization problems (CMOPs) are prevalent in practical applications, where constraints play a significant role. Building on techniques from constrained single-objective optimization, classical methods such as the constrained dominance principle have been extended to CMOPs. However, these methods struggle with CMOPs characterized by complex infeasible regions. Furthermore, as the number of decision variables increases, the search efficiency of algorithms deteriorates dramatically. To solve those issues, we propose a staged fuzzy evolutionary algorithm (i.e., SFEA) for constrained large-scale problems. To balance exploration and exploitation, a fuzzy stage adjustment strategy based on the sigmoid function is proposed. Furthermore, this article develops an improved fuzzy operator to perform fuzzy operations on various vectors (e.g., solutions or constraint violations). Computational experiments were conducted on CMOP test suites with up to 500 decision variables and a series of real-world applications. The experimental results demonstrate that, compared to existing peer algorithms, our algorithm exhibits superior or competitive performance.

1. Introduction

Constrained multiobjective optimization problems (CMOPs) frequently arise in practical applications, typically involving multiple conflicting objectives alongside one or more equality or inequality constraints. Examples include capacitated vehicle routing problems [1], dispatch problems [2], and biomarker identification problems [3], all of which can be modeled as CMOPs. Solving these problems requires algorithms that effectively balance objective optimization with constraint satisfaction.

Evolutionary algorithms possess the characteristic of utilizing populations to approximate optimal solutions, they portray a balance among multiple conflicting objectives. The effectiveness of evolution algorithms in solving MOPs has been confirmed [4]. In past studies, many multiobjective evolutionary algorithms (MOEAs) have emerged, including dominance-based [5], decomposition-based [6], and indicator-based MOEAs [7]. However, much of the early research was focused on unconstrained multiobjective optimization scenarios.

In contrast to MOPs, CMOPs pose increased complexity for solution methodologies. The heightened difficulty arises from the simultaneous optimization of multiple conflicting objectives, along with handling various constraints. Subject to various constraints, the entire search space is divided into two regions: the feasible and infeasible regions. The search region that satisfies all constraints is called the feasible region; otherwise, it is the infeasible region. The feasible regions can be extremely narrow or even discretely distributed. In the end, the difference between the constrained Pareto front (CPF) and the unconstrained Pareto front (UPF) regarding shape and position can be significant [8]. Overall, it is difficult for a constrained MOEA (CMOEA) to approximate the complete set of Pareto optimal solutions in a single run.

Based on single-objective optimization, constraint-handling techniques (CHTs), including constraint dominance principle (CDP) [5], methods based on penalty function [9], ϵ methods [10], methods based on multiobjective optimization [11], and hybrid methods [12] are developed for CMOPs. Unfortunately, since the infeasible regions of CMOPs may be very complex, resulting in CPFs can be discontinuous or even discrete. Classical CHTs encounter difficulties when dealing with these CMOPs, such as being easily stuck in local optimal regions or trapped in the partial global optimum regions. Considering the above issues, some studies have endeavored to enhance classical CHTs. These studies focus on preserving a portion of promising infeasible solutions, aiming to bridge the gap between the infeasible and feasible regions. For instance, approaches based on the angle CDP [13] and localized applications of CDP [8] have been explored. Recently, most of the popular CMOEAs use multi-population or multi-stage strategies and

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Nomenclature

CMOP	Constrained multiobjective optimization problem
MOEA	Multiobjective evolutionary algorithm
CMOEA	Constrained multiobjective evolutionary algorithm
CPF	Constrained Pareto front
UPF	Unconstrained Pareto front
CHT	Constraint-handling technique
CDP	Constraint dominance principle
IGD	Inverted generational distance
HV	Hypervolume
CSO	Competitive swarm optimization
DE	Differential evolution
PM	Polynomial mutation
SBX	Simulated binary crossover

show very promising performance. For multi-population-based methods, the focus is on utilizing multiple different populations to solve the original constrained optimization problem and auxiliary problems. For multi-stage-based methods, it emphasizes dividing the evolutionary process into multiple sub-stages and using different CHTs in different stages. Existing CMOEAs have demonstrated very good performance in solving CMOPs with small-scale decision variables. However, as the number of decision variables grows, the infeasible region also grows exponentially, leading to a drastic performance degradation of the algorithms.

Since fuzzy theory has the advantages of being easy to handle multi-source information and being flexible, designing fuzzy-based evolutionary algorithms is very promising. Wu et al. [14] are considered to have pioneered the introduction of a fuzzy penalty function approach for constrained optimization problems, where the penalty parameters are determined by a set of fuzzy rules. Yan et al. [15] proposed a fuzzy assignment scheme based on a bipartite graph for the assignment of customers and satellites, and it was applied to solve the two-echelon vehicle routing problems. Wang et al. [16] introduced an adaptive fuzzy constraint handling approach to adjust penalty coefficients, which are tuned at both the population and individual levels. However, the above methods only focus on single-objective optimization problems. Recently, several works have applied fuzzy theory to multiobjective evolutionary optimization. For instance, Yang et al. [17] divided the evolutionary course into fuzzy and precise stages and introduced a framework for the fuzzy decision variables for large-scale MOPs. Furthermore, Wang et al. [18] extended the framework of fuzzy decision variables based on linear search and the ratio of non-dominated solutions. Han et al. [19] proposed a fuzzy constraint processing method, which uses fuzzy set theory to characterize the differences between solutions based on objective function values and constraint violations.

Motivated by the considerations outlined above, we introduce a staged fuzzy evolutionary algorithm (SFEA) for constrained large-scale multiobjective optimization. Recognizing the critical importance of achieving a balance of exploration and exploitation, we underscore the necessity of dynamically adjusting the emphasis on these pivotal aspects. To this end, we propose a fuzzy stage adjustment strategy based on the sigmoid function. In the early stages, emphasizing the exploration of regions where optimal solutions may be located. In the later stages, focusing on exploring the optimal solutions themselves. Additionally, we devise a novel variant of the fuzzy operator to perform fuzzy operations on solutions vectors and constraint violation vectors.

The contributions of this work are as follows:

(1) We propose a CMOEA, termed as SFEA, for large-scale CMOPs. SFEA utilizes two distinct populations, each independently fuzzy evolving through the same optimizer. The populations approximate the CPF in both directions, from the feasible regions as well as the infeasible regions.
(2) To balance global exploration and local exploitation, a fuzzy stage adjusting strategy is designed. Based on the strategy, the evolutionary stage is adjusted into several fuzzy sub-stages. In the early stages, vectors are highly fuzzy, emphasizing global exploration. In the later stages, vectors are less fuzzy, focusing on local exploitation.
(3) To narrow down the search space, we propose a novel variant of the fuzzy operator. In contrast to existing fuzzy operators, it incorporates normalization and inverse normalization. The advantage of this variant lies in its ability to enhance the efficiency of the fuzzy operator in handling vectors with different scales, without compromising accuracy.
(4) To verify the effectiveness of the SFEA, computational experiments are conducted on several artificial test suites with up to 500 decision variables and 15 real-world application CMOPs. The results show that SFEA achieves more competitive performance compared to peer algorithms.

The remainder section is organized as follows. Section 2 introduces the background of CMOPs and fuzzy theory, followed by a literature review. The proposed SFEA is elaborated in Section 3. Then, Section 4 gives the experimental setup. Next, Section 5 presents experimental results and analysis. Finally, Section 6 concludes the work.

2. Related work

In this section, the basic definition of CMOPs as well as fuzzy theory are introduced. Subsequently, an overview of the literature on constrained multiobjective optimization is provided.

2.1. Definition of CMOPs

(1) *Constrained Multiobjective Optimization Problem:*

$$\min \mathbf{F}(\vec{x}) = \left(f_1(\vec{x}), f_2(\vec{x}), \ldots, f_M(\vec{x})\right),$$

$$s.t. \begin{cases} g_j(\vec{x}) \le 0, & j = 1, \ldots, p \\ h_j(\vec{x}) = 0, & j = p+1, \ldots, l \\ \vec{x} = (x_1, x_2, \ldots, x_D)^T \in \mathbb{R}^D, \end{cases} \tag{1}$$

where $\vec{x}$ denotes a D-dimensional decision vector, and $\mathbf{F}(\vec{x})$ is the objective vector with M conflicting objectives. The constraints $g_j(\vec{x}) \le 0$ and $h_j(\vec{x}) = 0$ represent p inequality constraints and l equality constraints, respectively.

(2) *Constraint Violation:* For a CMOP defined in Eq. (1), the violation of the jth constraint by a solution $\vec{x}$ can be defined as:

$$CV_j(\vec{x}) = \begin{cases} max(0, g_j(\vec{x})), & j = 1, \ldots, p, \\ max(0, |h_j(\vec{x})| - \eta), & j = p+1, \ldots, p+l. \end{cases} \tag{2}$$

where η is a small positive value that relaxes the equality constraints. The overall constraint violation for solution $\vec{x}$ is then calculated as:

$$CV(\vec{x}) = \sum_{j=1}^{p+l} CV_j(\vec{x}). \tag{3}$$

(3) *Feasible Region:* The feasible region, where $CV(\vec{x}) = 0$, is defined as:

$$\mathbb{FR} = \{\vec{x} \in \mathbb{R}^D | CV(\vec{x}) = 0\}. \tag{4}$$

2.2. Fuzzy theory

In general, the relationship between an element and a set is limited to "belonging" or "not belonging". Let U be the objects in the domain, and its general element is denoted by u; A is a set on the domain U. The relation between the domain U and the set A is defined as follows [20]:

$$f_A(u) = \begin{cases} 1, u \in A \\ 0, u \notin A \end{cases}, \forall u \in U \tag{5}$$

Fuzzy set theory serves as a natural extension of ordinary set theory, broadening the 0–1 value of the relationship function to the [0,1] interval. To this end, fuzzy membership function can be formalized as:

$$f_{\tilde{A}} : U \to [0,1], u \to f_{\tilde{A}}(u). \tag{6}$$

Based on the fuzzy membership function, a fuzzy set $\tilde{A}$ in U can be expressed by ordered pairs:

$$\tilde{A} = \left\{ \left(u, f_{\tilde{A}}(u) \mid u \in U \right) \right\}. \tag{7}$$

Here, $f_{\tilde{A}}$ represents the fuzzy membership function, and $f_{\tilde{A}}(u)$ denotes the fuzzy membership degree value. The membership degree value reflects the "grade of membership" of u to the fuzzy set $\tilde{A}$. Intuitively, a larger value with membership degree indicates that u has a higher membership grade in the fuzzy set. Concerning membership degree values, there are three cases: when $f_{\tilde{A}}(u) = 0$, the element u does not belong to the fuzzy set at all; when $f_{\tilde{A}}(u) = 1$, u entirely belongs to the fuzzy set; and when $f_{\tilde{A}}(u) \in [0,1]$, u partially belongs to the fuzzy set.

2.3. Literature review

In the early stages of constrained multiobjective evolutionary optimization, researchers primarily dedicated to developing CHTs. Specifically, popular CHTs fall into the following categories:

(1) *Constrained Dominance Principle* [5]: One of the most classical CHTs is the CDP, introduced by Deb et al. in 2012. In CDP, prioritizing feasible solutions over infeasible ones is the fundamental principle. More precisely, when comparing two solutions, $\vec{x}$ and $\vec{y}$, $\vec{x}$ is deemed superior to $\vec{y}$ under the following conditions: (1) if both $\vec{x}$ and $\vec{y}$ are feasible solutions, and $\vec{x} \prec \vec{y}$; (2) if both $\vec{x}$ and $\vec{y}$ are infeasible solutions, and $CV(\vec{x}) < CV(\vec{y})$; (3) if $\vec{x}$ is a feasible solution, and $\vec{y}$ is an infeasible solution. It can be observed that CDP exhibits a strong bias towards feasible solutions, thereby overlooking the potential role of infeasible solutions.

(2) *Methods Based on Penalty Function* [9]: In such methodologies, a penalty term representing the constraint violations of a solution is incorporated into the objective function. This conversion transforms CMOPs into unconstrained MOPs, allowing the application of canonical MOEAs. While these approaches are conceptually straightforward, tailoring appropriate penalty factors for diverse CMOPs proves to be a non-trivial task.

(3) *ϵ Methods* [10]: In order to exploit information in infeasible solutions, ϵ methods set a control parameter to relax the constraint boundaries. The methods are widely employed in addressing constrained optimization problems, particularly those with discontinuous feasible regions. Since promising infeasible solutions can serve as bridges connecting various discontinuous or even discrete feasible regions [21]. For instance, Zhou et al. [22] developed a CHT that emphasizes the diversity maintenance of infeasible solutions, achieving competitive results on CMOPs with discrete feasible regions. However, similar to penalty-based methods, it requires setting parameters to define the degree of constraint relaxation. The control parameter for the ϵ level will significantly impact the performance of the algorithms.

(4) *Methods Based on Multiobjective Optimization [11]:* Various constraints (i.e., equational or inequality constraints) are transformed into one or even multiple objectives for optimization. On the one hand, the advantage of this approach is that CMOPs, once transformed, can be directly solved using unconstrained MOEAs. On the other hand, assigning the same weights to the original objectives and the newly introduced objectives leads to a reduction in selection pressure, potentially resulting in the phenomenon of domination resistance [23].

(5) *Hybrid Methods* [12]: To achieve better performance, various popular CHTs are integrated in these methods, rather than relying solely on a single CHT. Nevertheless, exploring the advantages of each CHT for different CMOPs or at various stages of the optimization process remains a challenging task.

Many of the CHTs mentioned above are based on constrained single-objective evolutionary optimization. Therefore, to cope with more complex CMOP scenarios, further improvements are needed. Given the obvious bias that many CHTs (e.g., CDP [5] and ϵ methods [10]) exhibit towards feasible solutions, it is easy for algorithms to get stuck in local optimal regions. With this in mind, many research efforts have attempted to preserve some potential infeasible solutions. For instance, an angle-based CDP for CMOPs was designed by Fan et al. [13]. In the angle-based CDP, it utilized the angle information to preserve some infeasible solutions for maintaining population diversity. Zhou et al. [8] proposed a CMOEA, called MOEA/D-LCDP, applying CDP in a localized manner. In MOEA/D-LCDP, it capitalizes on the fast convergence speed of CDP, simultaneously addressing its susceptibility to local optima. Ma et al. [24] designed an algorithm, called NSGA-II-TOR, based on two rankings. In NSGA-II-TOR, the weights of constraint violations are adaptively increased during the course of evolution. However, the weights of the constraint violations are always greater than the objective functions, potentially resulting in underutilization of the information derived from the objectives. To address this concern, Yu et al. [25] suggested an algorithm based on dynamic preference selection assistance, assigning higher weights to the objective functions during the initial stages of evolution. Moreover, to strike a balance between constraints and objective functions, Zhou et al. [26] suggested a strictly constrained dominance relation along with a density estimation strategy based on shifts.

Some popular CMOEAs use multiple archives or populations, capitalizing on their collaboration to balance objectives and constraints. Wang et al. [27] proposed a differential evolution framework, which contains M sub-populations and an archive population, addressing M-constrained single-objective optimization problems and the CMOP with M objectives, respectively. However, the strong cooperation between these two populations imposes limitations on algorithm performance. To overcome the issue, Tian et al. [28] introduced a coevolutionary framework, named CCMO, with a focus on weak cooperation, yielding promising results. Since the second population dedicated to searching for UPF, CCMO does not deal well with CMOPs where CPF and UPF are very different. To this end, Zhou et al. [29] developed a dual-population algorithm for CMOPs, called DPACMO. In DPACMO, dual population work to approximate CPF from both sides of the constraint boundaries. Qiao et al. [30] developed a dynamic constraint processing mechanism and resource allocation scheme within a dual-population framework. He et al. [31] employed a dual-population strategy and proposed a self-organizing map approach for constrained multiobjective optimization. Zhou et al. [32] developed a competitive and cooperative swarm optimizer for large-scale CMOPs. The competitive swarm exhibits fast convergence characteristics, while the cooperative swarm is effective at overcoming local optima. Additionally, Qiao et al. [33] proposed a constraint separation method to identify promising individual constraints and leverage them to solve the original CMOPs. The original CMOP is thus transformed into a multitasking optimization problem, where $n+1$ populations are used to solve $n+1$ auxiliary tasks.

Recently, some studies have employed multi-stage optimization strategies to retain a portion of infeasible solutions at specific stages, demonstrating highly promising performance. In general, the evolutionary course is divided into multiple sequences or switchable stages, with an emphasis on objective optimization in the earlier stages. Xiang et al. [34] proposed a two-stage based algorithm called ToM based on the type of CMOPs. In ToM, the two stages prioritize the optimization of objective functions and constraints, respectively. Similarly, Liang et al. [35] suggested a CMOEA iterative process divided into learning and evolution stages. In the learning stage, the focus is on estimating the relationship between UPF and CPF, while in the evolution stage, the aim is to enhance the efficiency of utilizing objective functions. Yang et al. [36] proposed that search modes are divided into unconstrained and constrained optimization, dynamically switching between the two modes based on the population's state (i.e., the number of evolutionary generations and the proportion of feasible solutions). Tian et al. [37] developed a CMOEA with a two-stage fitness evaluation strategy, by switching the two stages to guide the population through infeasible regions and spread along the feasible boundaries. Karunanayake et al. [38] proposed a multi-level genetic algorithm that leverages solutions from one level to enhance optimization at another, resulting in a synergistic effect.

3. Staged fuzzy evolutionary algorithm

3.1. Main framework

The core concept of the proposed SFEA is that two populations utilize the same optimizer to approximate the true Pareto Front (CPF) from two distinct directions, while sharing offspring information during environmental selection. As a result, the populations approximate the CPF from both feasible and infeasible regions. Specifically, Pop_1 uses a fuzzy operator to fuzzify decision variables, thereby approximating the CPF from the feasible regions. Conversely, Pop_2 applies the fuzzy operator to both decision variables and constraint violations, estimating the true PF from the direction of infeasible regions.

Algorithm 1 Framework of SFEA

Input: N (population size), G (maximum generation)
Output: $Arch$ (archive set)
1: $Pop_1 \leftarrow$ Randomly initialize N solutions
2: $Pop_2 \leftarrow$ Randomly initialize N solutions
3: $CV_1 \leftarrow$ Calculate the CV for Pop_1
4: $CV_2 \leftarrow$ Calculate the CV for Pop_2
5: $g \leftarrow 0$
6: $i \leftarrow 0$
7: **while** $g \leq G$ **do**
8: $MatingPool_1 \leftarrow$ Binary_Tournament_Selection(Pop_1, N)
9: $MatingPool_2 \leftarrow$ Binary_Tournament_Selection(Pop_2, N)
10: $Off_1 \leftarrow$ Generate $N/2$ offspring using reproduction
11: operators based on parents from $MatingPool_1$
12: $Off_2 \leftarrow$ Generate $N/2$ offspring using reproduction
13: operators based on parents from $MatingPool_2$
14: $Off_1' \leftarrow$ Fuzzy_Operator (Off_1, i)
15: $[Off_2', CV_2'] \leftarrow$ Fuzzy_Operator($[Off_2, CV_2], i$)
16: $Q_1 \leftarrow Pop_1 \cup Off_1' \cup Off_2'$
17: $Q_2 \leftarrow Pop_2 \cup Off_2' \cup Off_1'$
18: $Pop_1 \leftarrow$ Environmental_Selection(Q_1, CV_1, N)
19: $Pop_2 \leftarrow$ Environmental_Selection(Q_2, CV_2', N)
20: $Arch \leftarrow$ Archive_Updating($Pop_1, Pop_2, Arch$)
21: $g \leftarrow g + 1$
22: $i \leftarrow$ Stage_Adjusting(g)
23: **end while**
24: **return** $Arch$

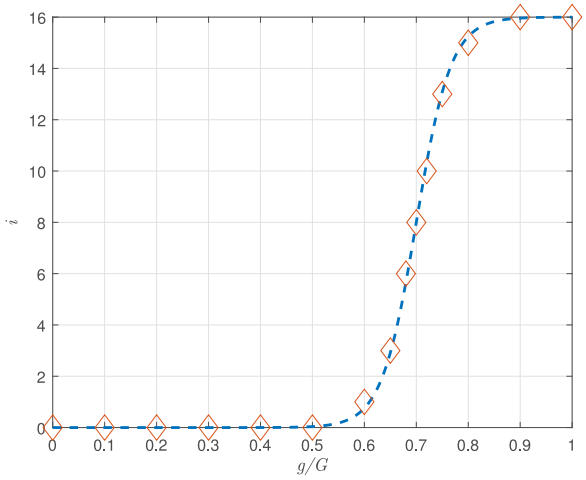


Fig. 1. Dynamic trajectory of the evolutionary stage i with iteration.

In Algorithm 1, we outline the framework of the SFEA. Initially, two populations (denoted as Pop_1 and Pop_2), each with N solutions, are randomly initialized (lines 1–2). Next, the constraint violations of the two populations are calculated (lines 3–4). Subsequently, the iterative process begins. We use the binary tournament strategy to select N solutions from each population to form $MatchingPool_1$ and $MatchingPool_2$ (lines 8–9). Based on the parents from the matching pool, $N/2$ new solutions are generated using reproduction operators, resulting in offspring populations denoted as Off_1 and Off_2 (lines 10–13). Then, the fuzzy operators are applied to Off_1 and Off_2, yielding the fuzzy offspring populations Off_1' and Off_2' (lines 14–15). Notably, for Off_1 the fuzzy operation acts only on the solutions, while for Off_2 it acts on both the solutions and their constraint violations. Following this, the two fuzzy offspring populations are combined with Pop_1 and Pop_2 to form Q_1 and Q_2, respectively. Next, in lines 18–19, the combined populations are performed environment selection. For the combined populations Q_1 and Q_2, along with their respective constraint violations, N solutions are selected using CDP [5] and the truncation operator from SPEA2 [39] to form the next generation of populations. Afterward, the $Arch$ is updated by the populations (line 20). Finally, the fuzzy evolution stage is adjusted (line 22).

3.2. Fuzzy stage adjusting strategy

To balance exploration and exploitation, an intuitive idea is to divide the evolutionary process into multiple evolutionary stages. With this in mind, this article introduces a novel adaptive strategy for adjusting the fuzzy stages.

Generally, exploration should be emphasized in the early stages of evolution, while exploitation should be emphasized in the later stages. To achieve this, we employ the sigmoid function for adjusting fuzzy stages, which is widely used in the evolutionary computing community [40]. Specifically, the mathematical formula for fuzzy evolutionary stage i is as follows:

$$i = \frac{S}{1 + e^{-30(g/G-\kappa)}}. \tag{8}$$

Here, g represents the current number of generations, and G denotes the maximum number of evolutionary generations. S indicates the maximum number of fuzzy evolutionary stages. The key parameter κ belongs to the interval [0, 1] to control the increasing trend of i. In this study, κ is set to 0.7 and S is set to 16. A detailed analysis of these parameters will be presented in Section 5.5.

As depicted in Fig. 1, the curve illustrates the trajectory of the fuzzy evolution stage i with increasing iteration counts. From the curve, it is

evident that in the early stages, i is relatively small, indicating a high degree of fuzziness, and the algorithm emphasizes global exploration. During these stages, although the fuzzy population cannot directly pinpoint the optimal solution, it can quickly approach the optimal solution, significantly reducing the search space. This capability is beneficial for exploring the regions where the optimal solution is located. Conversely, in the later stages, i is relatively large, indicating a low degree of fuzziness, and the algorithm focuses on exploitation. During these stages, the increasing precision of solutions motivates the population to pinpoint the optimal solution.

Overall, Eq. (8) is considered to be a suitable strategy for adjusting the fuzzy evolution stages. It features the ability to balance exploration and exploitation, thereby helping the population approximate the true PF.

3.3. Fuzzy operation

Based on Section 3.2, the entire evolutionary process is divided into several sub-stages. In this section, we will describe how the fuzzy operation is performed on the vectors in each sub-stage.

To reduce the search space, we devised an improved fuzzy operator to perform fuzzy operations on the original vector. The basic fuzzy theory required has been introduced in Section 2.2. Unlike the literature [17], we suggest a normalization operation and an inverse normalization operation for vectors. Specifically, we normalize the vectors before applying fuzzy operations. Subsequently, after the fuzzy operations, we conduct inverse normalization on the vectors. The advantage of normalization and inverse normalization operations is to enhance the efficiency of fuzzy operations on vectors with different scales.

Let u_n be the nth element of the original vector $\vec{u} = (u_1, u_2, \ldots, u_D)^T$, and u'_n be the element after fuzzy operation. The formulas of fuzzy operators are as follows:

$$u_n = \frac{u_n - z_n^*}{z_n^{nadir} - z_n^*}, \tag{9}$$

$$\Gamma_n^1 = \left\lfloor 10^i \cdot R_n^{-1} \cdot \left(u_n - u_n^l\right) \right\rfloor \cdot R_n \cdot 10^{-i} + u_n^l, \tag{10}$$

$$\Gamma_n^2 = \left\lceil 10^i \cdot R_n^{-1} \cdot \left(u_n - u_n^l\right) \right\rceil \cdot R_n \cdot 10^{-i} + u_n^l, \tag{11}$$

$$f_{\tilde{A}_1}(u_n) = \frac{1}{u_n - \Gamma_n^1}, \tag{12}$$

$$f_{\tilde{A}_2}(u_n) = \frac{1}{\Gamma_n^2 - u_n}, \tag{13}$$

$$\tilde{A}_1 = \left\{ \left(u_n, f_{\tilde{A}_1}\left(u_n\right) \mid n = 1, 2, \ldots, D \right) \right\}, \tag{14}$$

$$\tilde{A}_2 = \left\{ \left(u_n, f_{\tilde{A}_2}\left(u_n\right) \mid n = 1, 2, \ldots, D \right) \right\}, \tag{15}$$

$$u'_n = \begin{cases} \Gamma_n^1, f_{\tilde{A}_1}\left(u_n\right) > f_{\tilde{A}_2}\left(u_n\right) \\ \Gamma_n^2, f_{\tilde{A}_1}\left(u_n\right) < f_{\tilde{A}_2}\left(u_n\right) \\ \text{rand}\left(\Gamma_n^1, \Gamma_n^2\right), f_{\tilde{A}_1}\left(u_n\right) = f_{\tilde{A}_2}\left(u_n\right) \end{cases}, \tag{16}$$

$$u'_n = u'_n * (z_n^{nadir} - z_n^*) + z_n^*, \tag{17}$$

where z^* denotes the utopia vector, while z^{nadir} represents the nadir vector. u_n^l and R_n are the lower bound and value range of the nth element. i denotes the ith fuzzy evolutionary stage. Γ_n^1 and Γ_n^2 are the possible binary values of the nth element after fuzzy. $f_{\tilde{A}_1}$ and $f_{\tilde{A}_2}$ are fuzzy membership functions; accordingly, $\tilde{A}_1$ and $\tilde{A}_2$ are fuzzy sets. Eq. (9) represents the normalization of the original vector. Eqs. (10)–(16) calculate the fuzzy membership function values for u_n, and fuzzy u_n into a fuzzy target value with a large membership degree. Eq. (17) denotes the inverse normalization of the fuzzy vector.

Algorithm 2 elaborates on the details of the fuzzy operator. Firstly, the original vector is normalized. Secondly, determine the range of the element based on its upper and lower bounds. Next, calculate the fuzzy target value of vector $\vec{u}$ (lines 3–4). Then, the membership degree of the original vector in the fuzzy set is calculated (lines 5–6). Afterward, the values of the original vector are updated to the fuzzy target values with large membership values (lines 7–13). Finally, the fuzzy vector is inverse normalized.

Algorithm 2 Fuzzy__Operation $(\vec{u}, i)$

Input: $\vec{u}$ (original vector), i (i-th fuzzy stage)
Output: $\vec{u}'$ (fuzzy vector)
1: $\vec{u} \leftarrow (\vec{u} - z^*)/(z^{nadir} - z^*)$
2: $R \leftarrow \vec{u}_u - \vec{u}_l$
3: $\Gamma^1 \leftarrow 10^{-i} * R * floor(10^i * R^{-1} * (\vec{u} - \vec{u}_l)) + \vec{u}_l$
4: $\Gamma^2 \leftarrow 10^{-i} * R * ceil(10^i * R^{-1} * (\vec{u} - \vec{u}_l)) + \vec{u}_l$
5: $f_{\tilde{A}_1} \leftarrow 1/(\vec{u} - \Gamma^1)$
6: $f_{\tilde{A}_2} \leftarrow 1/(\Gamma^2 - \vec{u})$
7: **if** $f_{\tilde{A}_1} > f_{\tilde{A}_2}$ **then**
8: $bool \leftarrow true$
9: **else**
10: $bool \leftarrow false$
11: **end if**
12: $\vec{u}' \leftarrow \Gamma^2$
13: $\vec{u}'(find(bool)) \leftarrow \Gamma^1(find(bool))$
14: $\vec{u}' \leftarrow \vec{u}' * (z^{nadir} - z^*) + z^*$
15: **return** $\vec{u}'$

In general, the fuzzy operation can fuzzy the vectors to different degrees based on the fuzzy sub-stage, thereby reducing the range of the search space.

3.4. Computational complexity of SFEA

Based on the framework described in Algorithm 1, the proposed SFEA algorithm consists of the following key components: Binary_Tournament_Selection(), Generate_Offspring(), Fuzzy_Operator(), Environmental_Selection(), and Archive_Updating(). Let N represent the population size, M the number of objectives, and D the number of decision variables. In the worst-case scenario, the computational complexities of these components are $O(N)$, $O(ND)$, $O(ND)$, $O(N^3)$, $O(MN^2)$, and $O(N)$, respectively. Therefore, the overall computational complexity of the proposed SFEA is $O(N^3)$.

4. Experimental SETUP

4.1. Test suites

To demonstrate the effectiveness of the SFEA, we selected four benchmark-constrained test suites, including constrained DTLZ (i.e., C-DTLZ [41] and DC-DTLZ [42]), DAS-CMOPs [43], and DOC [44].

The constrained DTLZ suite includes ten test instances of CMOPs. The C-DTLZ problems are relatively straightforward, with challenges arising from regular and irregular CPF shapes. In contrast, the DC-DTLZ problems are characterized by discontinuous CPFs and constraints in the decision space, which make CMOEAs prone to falling into local optima. For the DAS-CMOPs suite, it comprises nine CMOPs, each presenting challenges in terms of convergence, diversity, and feasibility. Constrained DTLZ and DAS-CMOPs are scalable suites for the number of decision variables. The DOC suite includes nine instances with various equality and inequality constraints in both the decision and objective spaces. Although not scalable, it represents one of the most challenging CMOP suites for CMOEAs. Additionally, to assess the practical performance of CMOEAs, 15 CMOPs derived from real-world applications [45] are used for verification.

4.2. Performance indicator

Different from inverted generational distance (IGD) [46] is Pareto non-compliant, IGD$^+$ indicator [47] is weakly Pareto compliant. To this end, for CMOPs where the true PF is known, the IGD$^+$ [47] is adopted to evaluate the performance of CMOEAs. However, in practical applications where the CPF is unknown, we resort to the hypervolume (HV) indicator [48]. Since the HV indicator can simultaneously evaluate the convergence and diversity of the solution set without the need for uniformly distributed reference points on the CPF.

Specifically, the IGD$^+$ is defined as follows:

$$IGD^+(P) = \frac{1}{|Z|}\sum_{j=1}^{|Z|} d_j^+(\vec{z},\vec{x}), \tag{18}$$

where $\vec{z} \in Z$, $\vec{x} \in P$ and $d^+(\vec{z},\vec{x})$ is defined as follows:

$$d^+(\vec{z},\vec{x}) = \sqrt{\left(\max\left\{x_1 - z_1, 0\right\}\right)^2 + \ldots + \left(\max\left\{x_M - z_M, 0\right\}\right)^2}. \tag{19}$$

The HV metric is defined as follows:

$$HV(P) = \mathrm{VOL}(\bigcup_{\vec{x}\in P} [x_1, z_1^r], \ldots, [x_M, z_M^r]), \tag{20}$$

where VOL denotes the hypervolume about the nadir point $z^r = (z_1^r, \ldots, z_M^r)^T$.

In addition, CMOEAs are independently run 25 times on all CMOPs. The mean and standard deviation regarding the IGD$^+$ or HV indicators are recorded.

4.3. Comparing algorithms

To verify the performance of the SFEA, this study selected six advanced algorithms, including POCEA [49], LMOCSO [50], MOEA/D-FCHT [19], CCMO [28], DPCPRA [30], and CMOSMA [31] for experimental comparison. Furthermore, all algorithms are implemented in PlatEMO [51].

(1) *POCEA* [49] (2021): An algorithm tailored for constrained large-scale multiobjective problems, emphasizing the importance of generating promising feasible solutions or valuable infeasible solutions.
(2) *LMOCSO* [50] (2020): A competitive swarm optimization (CSO) algorithm designed for large-scale multiobjective optimization incorporates a two-stage particle update strategy, markedly enhancing search efficiency.
(3) *MOEA/D-FCHT* [19] (2022): A fuzzy CHT is introduced, utilizing fuzzy set theory to characterize the differences among individual solutions with respect to objective values and constraint violations.
(4) *CCMO* [28] (2021): A framework based on two population co-evolutionary optimization that solves complex original CMOPs with the help of simple auxiliary problems and performs promisingly on a large number of CMOPs.
(5) *DPCPRA* [30] (2024): A dual-population evolutionary algorithm based on dynamic constraint handling and resource allocation is introduced, effectively balancing the search capabilities for constraints and objective functions.
(6) *CMOSMA* [31] (2022): A novel self-organizing mapping method is proposed, which, in conjunction with dual-population evolutionary strategies, significantly improves the ability to solve CMOPs with complex feasible regions.

4.4. Reproduction operators

In the experiments, both POCEA and LMOCSO adopt CSO as the reproduction operator. MOEA/D-FCHT uses differential evolution (DE) [52] operator and polynomial mutation (PM) [53] to generate offspring solutions. Regarding the DE operator, the control parameters CR and F are set to 1.0 and 0.5 respectively. The distribution index in PM is 20 while the mutation probability is $1/D$. CCMO employs the GA operator to generate the offspring population. The GA operator consists of two components: simulated binary crossover (SBX) [54] and PM. For SBX, its distribution index is 20 and the crossover probability is $1/D$.

Table 1
Settings on CMOPs regarding the number of objectives, population size, number of decision variables, and maximum number of evaluations.

Problems	M	N	D	E_{max}
Constrained DTLZ	2	100	100	80,000
			200	100,000
	3	150	500	300,000
DAS-CMOP1–6	2	100	100	100,000
			200	250,000
DAS-CMOP7–9	3	150	300	500,000
DOC1–7	2	100	6	20,0000
			16	
			10	
			8	
			8	
			11	
			11	
DOC8–9	3	150	10	
			11	

4.5. Other parameter settings

For the problems in test suites, Table 1 summarizes the specific settings. Here, M denotes the number of optimization objectives; N is the population size; D represents the decision variable dimension; and E_{max} is the maximum number of evaluations. Additionally, for CMOPs in real-world applications, the E_{max} is set to 200,000 for the algorithms to converge.

5. Results and analysis

5.1. Constrained DTLZ

Table 2 shows the mean and standard deviation results on the constrained DTLZ suite for all test instances regarding the IGD$^+$ indicator. The best results among the seven algorithms are highlighted. The terms "+", "−", and "≈" suggest that the peer algorithm is statistically better, worse, or no difference at a 5% significance level.

Concerning the number of highlighted best results, SFEA stands out as the best-performing, achieving the best results in 18 out of 24 instances of the C-DTLZ suite. Regarding CMOSMA, it achieved the best results in five out of six instances of C3-DTLZ4, indicating that the algorithm is highly suitable for addressing test problems with high diversity requirements. For CCMO, it only obtained three best results in instances of C3-DTLZ4. For other remaining CMOEAs, DPCPRA, MOEA/D-FCHT, POCEA and LMOCSO did not obtained any best results. Overall, for the C-DTLZ1 test suite with simple constraints, the proposed SFEA algorithm demonstrates excellent performance. The superior performance of SFEA can be attributed to the fuzziness of decision variables, which narrows down the search space of the populations and enhances the computational efficiency.

For better visualization, Fig. 2 illustrates the final solution sets obtained by seven algorithms on C1-DTLZ3 with 500 decision variables and two optimization objectives. Regarding convergence, the final solution sets obtained by six peer algorithms with objective values greater than a few hundred, indicating that they are unconverged. In contrast, the final solution set of SFEA converges both objectives to the 0–1 interval, demonstrating excellent convergence. Concerning diversity, POCEA exhibits the poorest performance, while LMOCSO,

Table 2
IGD$^+$ results achieved by POCEA, LMOCSO, MOEA/D-FCHT, CCMO, DPCPRA, CMOSMA, and SFEA on C-DTLZ test suite.

Problem	M	D	POCEA	LMOCSO	MOEA/D-FCHT	CCMO	DPCPRA	CMOSMA	SFEA
C1-DTLZ1	2	100	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.2912e−3 (4.96e−6)**
		200	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.2909e−3 (6.61e−6)**
		500	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.2911e−3 (7.85e−6)**
	3	100	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.1228e−2 (5.88e−5)**
		200	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.1210e−2 (5.92e−5)**
		500	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.1201e−2 (5.20e−5)**
C1-DTLZ3	2	100	1.3223e+2 (3.85e+1) −	1.0105e+2 (5.11e+1) −	7.0860e+1 (2.57e+1) −	2.4557e+1 (4.29e+0) −	2.3183e+1 (3.04e+0) −	3.7074e+1 (6.16e+0) −	**1.7453e−3 (1.29e−5)**
		200	2.6290e+2 (8.55e+1) −	1.7798e+2 (7.91e+1) −	1.4535e+2 (6.34e+1) −	9.5400e+1 (9.05e+0) −	9.4768e+1 (8.36e+0) −	1.1635e+2 (9.14e+0) −	**1.7532e−3 (1.81e−5)**
		500	6.5300e+2 (2.34e+2) −	3.9785e+2 (1.58e+2) −	3.8155e+2 (1.90e+2) −	2.6222e+2 (1.52e+1) −	2.4762e+2 (1.42e+1) −	2.9590e+2 (1.20e+1) −	**1.7453e−3 (1.40e−5)**
	3	100	9.6937e+1 (2.97e+1) −	9.0328e+1 (3.38e+1) −	4.3160e+1 (3.04e+1) −	4.1411e+1 (4.26e+0) −	4.3523e+1 (5.68e+0) −	6.5213e+1 (7.16e+0) −	**1.7956e−2 (2.59e−4)**
		200	2.2907e+2 (5.44e+1) −	1.8543e+2 (5.94e+1) −	1.2643e+2 (6.46e+1) −	1.3310e+2 (1.04e+1) −	1.3455e+2 (1.23e+1) −	1.8186e+2 (1.36e+1) −	**1.7885e−2 (2.08e−4)**
		500	4.6136e+2 (9.95e+1) −	3.5705e+2 (9.02e+1) −	3.6587e+2 (1.86e+2) −	3.7553e+2 (1.92e+1) −	3.7246e+2 (1.40e+1) −	4.2387e+2 (1.98e+1) −	**1.8040e−2 (3.86e−4)**
C2-DTLZ2	2	100	2.1871e−2 (2.09e−2) −	6.7158e−3 (5.76e−4) −	5.6150e−3 (3.48e−4) −	2.5062e−3 (8.40e−5) −	2.3554e−3 (7.74e−5) −	3.1221e−3 (1.31e−4) −	**1.9053e−3 (4.88e−5)**
		200	3.8655e−2 (3.30e−2) −	2.2851e−2 (2.00e−3) −	1.6834e−2 (1.55e−3) −	5.3692e−3 (3.83e−4) −	4.6158e−3 (2.66e−4) −	7.8264e−3 (7.08e−4) −	**1.9080e−3 (3.66e−5)**
		500	4.1105e−2 (3.35e−2) −	3.9351e−2 (2.66e−3) −	2.8807e−2 (2.27e−3) −	7.3203e−3 (3.64e−4) −	5.7263e−3 (3.36e−4) −	9.2362e−3 (6.76e−4) −	**1.9161e−3 (4.33e−5)**
	3	100	5.9530e−2 (2.31e−2) −	8.0041e−2 (5.74e−3) −	7.9277e−2 (7.85e−3) −	2.5483e−2 (1.05e−3) −	2.1849e−2 (8.95e−4) −	3.5786e−2 (2.05e−3) −	**1.5395e−2 (4.59e−4)**
		200	8.9246e−2 (2.84e−2) −	1.9850e−1 (1.75e−2) −	1.6497e−1 (1.55e−2) −	5.0217e−2 (3.21e−3) −	4.0428e−2 (2.94e−3) −	8.2111e−2 (4.94e−3) −	**1.5307e−2 (4.82e−4)**
		500	9.4836e−2 (2.40e−2) −	2.5802e−1 (1.76e−2) −	2.1511e−1 (1.39e−2) −	6.3638e−2 (3.19e−3) −	3.8062e−2 (5.39e−3) −	8.4107e−2 (4.85e−3) −	**1.5412e−2 (4.38e−4)**
C3-DTLZ4	2	100	2.0624e−2 (3.32e−3) −	1.0308e+0 (9.67e−2) −	8.8385e−3 (7.29e−4) −	**5.2757e−3 (1.37e−4)** +	4.2873e−2 (1.89e−1) −	8.0903e−2 (2.61e−1) −	7.7963e−3 (7.23e−4)
		200	2.0872e−2 (2.82e−3) −	1.0184e+0 (8.06e−2) −	1.0621e−2 (3.93e−3) ≈	1.5618e−1 (3.53e−1) −	8.0554e−2 (2.62e−1) −	**5.0421e−3 (1.37e−4)** +	9.0874e−3 (1.26e−3)
		500	1.7417e−2 (2.17e−3) −	9.8461e−1 (2.24e−2) −	8.5742e−2 (2.60e−1) −	4.2444e−2 (1.89e−1) −	5.0811e−3 (2.17e−4) +	**4.6099e−3 (1.17e−4)** +	5.7653e−3 (4.31e−4)
	3	100	1.7438e−1 (1.57e−1) −	9.0028e−1 (5.86e−1) −	9.9397e−2 (4.52e−2) −	5.8779e−2 (6.83e−2) −	4.5590e−2 (1.27e−3) +	**4.2201e−2 (1.06e−3)** +	5.6926e−2 (3.96e−3)
		200	1.9151e−1 (1.74e−1) ≈	5.0777e−1 (4.50e−1) −	2.0578e−1 (1.05e−1) −	5.8802e−2 (6.82e−2) +	5.9685e−2 (6.80e−2) +	**4.1730e−2 (1.04e−3)** +	6.7778e−2 (6.03e−3)
		500	1.8462e−1 (1.72e−1) −	4.8966e−1 (3.92e−1) −	3.3964e−1 (1.62e−1) −	4.4424e−2 (1.56e−3) +	5.2531e−2 (1.89e−3) ≈	**4.1616e−2 (1.12e−3)** +	5.1969e−2 (2.40e−3)
+/ − / ≈			0/23/1	0/34/0	0/23/1	3/21/0	3/20/1	5/19/0	

Table 3
IGD$^+$ results achieved by POCEA, LMOCSO, MOEA/D-FCHT, CCMO, DPCPRA, CMOSMA, and SFEA on DC-DTLZ test suite.

Problem	M	D	POCEA	LMOCSO	MOEA/D-FCHT	CCMO	DPCPRA	CMOSMA	SFEA
DC1-DTLZ1	2	100	6.9761e+2 (8.19e+1) −	4.0808e+2 (1.80e+2) −	2.2721e+2 (1.73e+2) −	1.1216e+2 (1.43e+1) −	9.8337e+1 (1.77e+1) −	1.6107e+2 (2.28e+1) −	**4.3689e−4 (3.64e−6)**
		200	1.6075e+3 (3.38e+2) −	8.0614e+2 (3.17e+2) −	6.2900e+2 (3.05e+2) −	4.0187e+2 (3.66e+1) −	3.4843e+2 (2.95e+1) −	4.6128e+2 (5.35e+1) −	**5.3127e−4 (4.54e−4)**
		500	4.5955e+3 (1.18e+3) −	1.7378e+3 (3.39e+2) −	1.6208e+3 (6.83e+2) −	1.0760e+3 (5.19e+1) −	8.7993e+2 (4.27e+1) −	1.1148e+3 (5.29e+1) −	**1.5515e−2 (7.54e−2)**
	3	100	5.0006e+2 (9.18e+1) −	3.5917e+2 (9.47e+1) −	1.5791e+2 (1.09e+2) −	1.4873e+2 (1.52e+1) −	1.5022e+2 (1.83e+1) −	2.1923e+2 (2.83e+1) −	**6.3362e−3 (4.14e−5)**
		200	1.2885e+3 (1.98e+2) −	6.3838e+2 (1.61e+2) −	5.3819e+2 (1.89e+2) −	4.7402e+2 (3.91e+1) −	4.2453e+2 (3.58e+1) −	6.6683e+2 (3.73e+1) −	**6.3338e−3 (4.47e−5)**
		500	2.7558e+3 (8.89e+2) −	1.4734e+3 (2.56e+2) −	1.2179e+3 (5.94e+2) −	1.2820e+3 (6.74e+1) −	1.1216e+3 (5.07e+1) −	1.4178e+3 (7.13e+1) −	**6.3216e−3 (3.92e−5)**
DC1-DTLZ3	2	100	1.5488e+2 (4.87e+1) −	1.0183e+2 (4.69e+1) −	8.7952e+1 (3.29e+1) −	2.8883e+1 (4.69e+0) −	2.6085e+1 (3.82e+0) −	4.2375e+1 (6.02e+0) −	**5.6029e−4 (5.86e−6)**
		200	3.6225e+2 (7.49e+1) −	1.8044e+2 (6.14e+1) −	1.2407e+2 (7.76e+1) −	1.0253e+2 (8.97e+0) −	9.5405e+1 (9.79e+0) −	1.2571e+2 (8.05e+0) −	**8.5309e−3 (2.75e−2)**
		500	7.7282e+2 (2.55e+2) −	3.6254e+2 (1.39e+2) −	3.7391e+2 (2.46e+2) −	2.6969e+2 (1.25e+1) −	2.1625e+2 (1.01e+1) −	3.0345e+2 (1.57e+1) −	**5.6080e−4 (6.05e−6)**
	3	100	1.4013e+2 (3.46e+1) −	1.0084e+2 (2.44e+1) −	5.4342e+1 (3.57e+1) −	4.4933e+1 (5.81e+0) −	4.5162e+1 (6.92e+0) −	6.3780e+1 (6.42e+0) −	**9.8914e−3 (1.79e−4)**
		200	2.7927e+2 (1.08e+2) −	1.8718e+2 (5.32e+1) −	1.3613e+2 (6.69e+1) −	1.3762e+2 (1.15e+1) −	1.3274e+2 (8.80e+0) −	1.8532e+2 (1.31e+1) −	**9.7793e−3 (2.20e−4)**
		500	5.1422e+2 (7.71e+1) −	3.8293e+2 (8.17e+1) −	4.0945e+2 (1.86e+2) −	3.9007e+2 (2.25e+1) −	3.7096e+2 (1.46e+1) −	4.3293e+2 (1.83e+1) −	**9.9144e−3 (1.56e−4)**
DC2-DTLZ1	2	100	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.2937e−3 (6.58e−6)**
		200	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.2921e−3 (6.00e−6)**
		500	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.2894e−3 (6.54e−6)**
	3	100	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.1242e−2 (5.66e−5)**
		200	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.1275e−2 (8.58e−5)**
		500	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.1220e−2 (5.02e−5)**
DC2-DTLZ3	2	100	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.7442e−3 (1.53e−5)**
		200	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.7450e−3 (1.49e−5)**
		500	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.7493e−3 (1.91e−5)**
	3	100	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.7814e−2 (3.19e−4)**
		200	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.7755e−2 (2.67e−4)**
		500	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.7840e−2 (3.01e−4)**
DC3-DTLZ1	2	100	7.8162e+2 (1.44e+2) −	3.9773e+2 (1.45e+2) −	2.1631e+2 (1.43e+2) −	1.5973e+2 (2.24e+1) −	1.7482e+2 (2.30e+1) −	2.0196e+2 (2.67e+1) −	**4.3832e−4 (3.49e−6)**
		200	1.7194e+3 (2.65e+2) −	8.2770e+2 (3.52e+2) −	5.1378e+2 (2.63e+2) −	4.6953e+2 (3.22e+1) −	4.9809e+2 (4.91e+1) −	5.4014e+2 (4.55e+1) −	**3.7723e−2 (1.52e−1)**
		500	4.4291e+3 (6.55e+2) −	1.8839e+3 (6.50e+2) −	1.6214e+3 (6.88e+2) −	1.2487e+3 (5.43e+1) −	1.4025e+3 (5.27e+1) −	1.3014e+3 (6.13e+1) −	**1.8947e−2 (9.26e−2)**
	3	100	6.8085e+2 (1.36e+2) −	4.2559e+2 (1.88e+2) −	1.6499e+2 (1.06e+2) −	3.1580e+2 (4.29e+2) −	2.5632e+2 (3.88e+1) −	3.4024e+2 (2.84e+1) −	**3.9374e−3 (6.28e−5)**
		200	1.4542e+3 (2.33e+2) −	7.9601e+2 (2.65e+2) −	4.3817e+2 (2.90e+2) −	9.0027e+2 (1.61e+2) −	1.0452e+3 (1.85e+2) −	1.1208e+3 (9.27e+1) −	**3.9222e−1 (4.04e−1)**
		500	3.6060e+3 (6.48e+2) −	1.8488e+3 (6.81e+2) −	1.5470e+3 (5.73e+2) −	1.5919e+3 (1.52e+2) −	1.6294e+3 (6.98e+1) −	1.9523e+3 (8.70e+1) −	**7.1732e−2 (1.64e−1)**
DC3-DTLZ3	2	100	1.8739e+2 (5.32e+1) −	1.0402e+2 (6.12e+1) −	5.5020e+1 (2.71e+1) −	3.8794e+1 (4.51e+0) −	4.9232e+1 (6.66e+0) −	4.9816e+1 (5.27e+0) −	**5.6128e−4 (5.89e−6)**
		200	3.6287e+2 (8.97e+1) −	1.7306e+2 (8.17e+1) −	1.3384e+2 (6.47e+1) −	1.2337e+2 (1.05e+1) −	1.3929e+2 (1.43e+1) −	1.4249e+2 (9.70e+0) −	**4.5385e−3 (1.99e−2)**
		500	9.0267e+2 (2.35e+2) −	4.2137e+2 (1.81e+2) −	3.4267e+2 (2.08e+2) −	3.1823e+2 (1.51e+1) −	3.6517e+2 (1.87e+1) −	3.4052e+2 (1.89e+1) −	**4.5360e−3 (1.99e−2)**
	3	100	1.9311e+2 (4.90e+1) −	1.1242e+2 (2.78e+1) −	5.9880e+1 (3.21e+1) −	5.3291e+1 (5.38e+0) −	5.6025e+1 (1.04e+1) −	9.4482e+1 (1.06e+1) −	**5.8354e−3 (1.56e−4)**
		200	3.8346e+2 (9.07e+1) −	2.5323e+2 (5.24e+1) −	1.5735e+2 (7.60e+1) −	1.6712e+2 (1.42e+1) −	1.7027e+2 (1.49e+1) −	2.7148e+2 (2.68e+1) −	**1.4730e−1 (2.13e−1)**
		500	9.1106e+2 (1.28e+2) −	4.5845e+2 (1.16e+2) −	3.4744e+2 (1.76e+2) −	4.2785e+2 (1.68e+1) −	4.6471e+2 (1.64e+1) −	5.1373e+2 (1.66e+1) −	**1.7147e−2 (3.17e−2)**
+/ − / ≈			0/36/0	0/36/0	0/36/0	0/36/0	0/36/0	0/36/0	

MOEA/D-FCHT, CCMO, DPCPRA, and CMOSMA show very poor convergence performance. Only the solution set of SFEA uniformly and comprehensively covers the CPF.

Different from the C-DTLZ test suite, the DC-DTLZ suite is more complex by introducing constraints in both decision and objective spaces. Table 3 shows the results of the seven algorithms on the DC-DTLZ test suite regarding the IGD$^+$ indicator. The proposed SFEA exhibits significant superiority over other compared algorithms, and it obtains the best performance results in all instances of the DC-DTLZ suite. Furthermore, the final populations with the median IGD$^+$ obtained by the seven algorithms on DC1-DTLZ1 with 500 decision variables and three objectives are presented in Fig. 3. Similarly, only the proposed SFEA converges to the true PF. The possible reason is that the fuzzy operator in SFEA is fuzzy for constraint violations and decision vectors. In contrast, the other six compared CMOEAs did not obtain any best results, not even any feasible solutions in DC2-DTLZ1 and DC2-DTLZ3. This is because the constraints in these instances make the reachable objective space infeasible, and both DTLZ1 and DTLZ3 are baseline MOPs with convergence-hard properties, which causes great difficulties for CMOEAs to converge to the CPF. Additionally, to visually assess the convergence speed of the algorithms, the convergence profiles of the seven algorithms regarding the IGD$^+$ values on C1-DTLZ3 with $M = 2, D = 500$, and DC1-DTLZ1 with $M = 3, D = 500$ are shown in Fig. 4.

5.2. DAS-CMOPs

The results regarding the IGD$^+$ indicator of each algorithm on the DAS-CMOPs suite are recorded in Table 4. On the whole, the proposed SFEA is the algorithm that achieves the best performance.

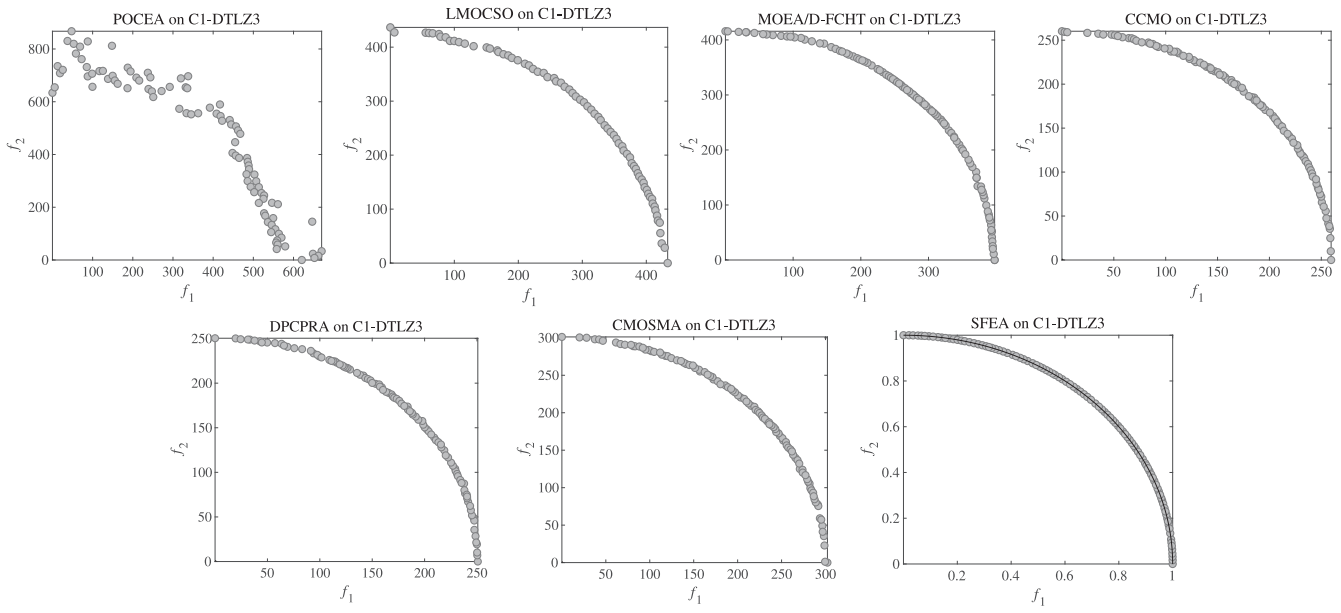


Fig. 2. Final solution set with median IGD$^+$ value among 25 runs of seven algorithms on C1-DTLZ3 with $M = 2, D = 500$.

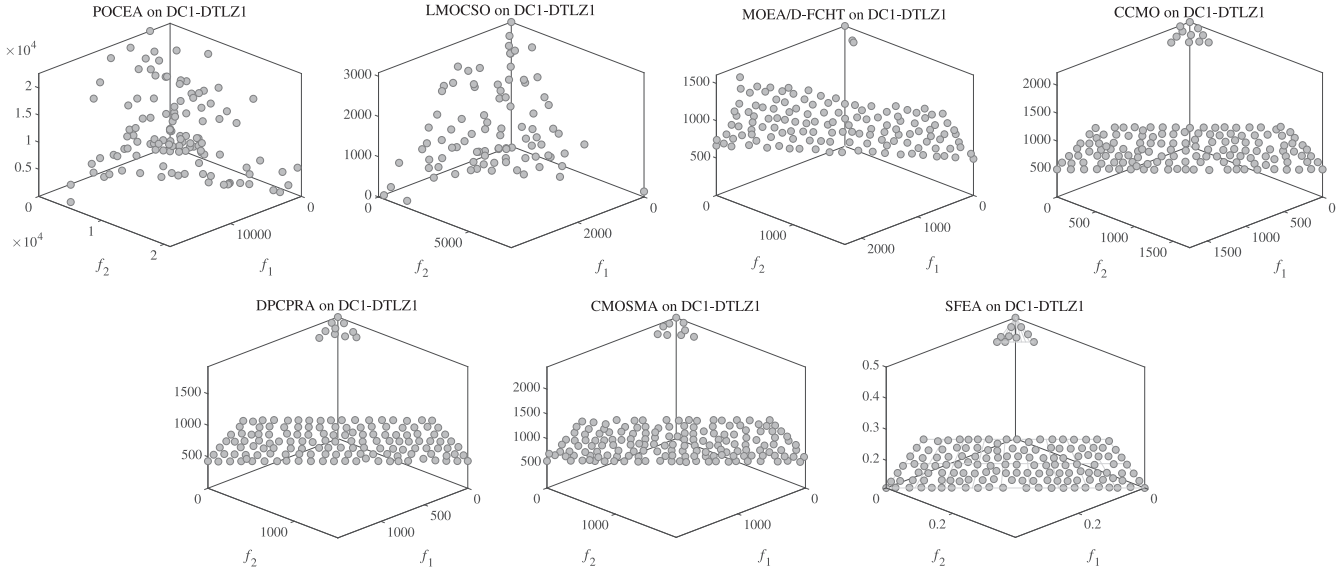


Fig. 3. Final solution set with median IGD$^+$ value among 25 runs of seven algorithms on DC1-DTLZ1 with $M = 3, D = 500$.

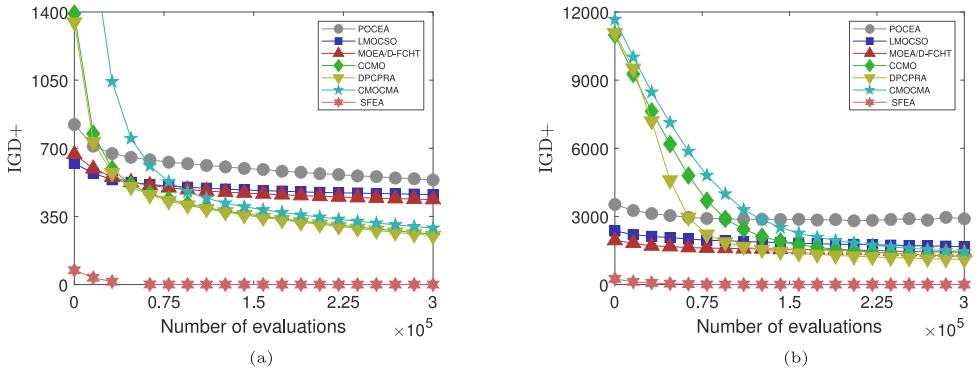


Fig. 4. Convergence profiles of the mean IGD$^+$ values for the seven algorithms over 25 independent runs. (a) C1-DTLZ3 with $M = 2, D = 500$. (b) DC1-DTLZ1 with $M = 3, D = 500$.

Table 4
IGD$^+$ results achieved by POCEA, LMOCSO, MOEA/D-FCHT, CCMO, DPCPRA, CMOSMA, and SFEA on DAS-CMOPs test suite.

Problem	M	D	POCEA	LMOCSO	MOEA/D-FCHT	CCMO	DPCPRA	CMOSMA	SFEA
DAS-CMOP1	2	100	1.3093e−1 (1.71e−1) −	NaN (NaN)	7.9026e−1 (5.84e−3) −	7.8147e−1 (2.16e−2) −	7.9234e−1 (1.74e−2) −	7.8458e−1 (1.83e−2) −	**3.2717e−3 (4.85e−4)**
		200	4.0743e−1 (1.52e−1) −	NaN (NaN)	8.0243e−1 (3.68e−3) −	7.9230e−1 (1.18e−2) −	8.0120e−1 (1.35e−2) −	8.0127e−1 (1.29e−2) −	**3.1012e−3 (9.84e−4)**
		300	7.2064e−1 (7.28e−2) −	NaN (NaN)	8.0690e−1 (4.55e−3) −	7.9996e−1 (9.37e−3) −	8.1245e−1 (1.13e−2) −	8.0638e−1 (1.03e−2) −	**3.1954e−3 (9.03e−4)**
DAS-CMOP2	2	100	3.7143e−2 (6.77e−3) −	NaN (NaN)	1.7751e−1 (2.15e−3) −	1.7312e−1 (3.24e−3) −	1.7513e−1 (1.93e−3) −	1.7361e−1 (4.09e−3) −	**5.2107e−3 (6.16e−4)**
		200	4.4714e−2 (8.76e−3) −	NaN (NaN)	1.8566e−1 (1.53e−3) −	1.8342e−1 (4.64e−3) −	2.8549e−1 (2.29e−1) −	1.8466e−1 (4.04e−3) −	**4.4031e−3 (1.30e−3)**
		300	1.2299e−1 (1.99e−2) −	NaN (NaN)	1.8897e−1 (1.33e−3) −	1.8782e−1 (4.46e−3) −	6.0277e−1 (2.85e−1) −	1.8948e−1 (4.84e−3) −	**3.7894e−3 (4.09e−4)**
DAS-CMOP3	2	100	1.8106e−1 (4.50e−2) −	NaN (NaN)	3.2863e−1 (1.68e−1) −	2.7866e−1 (2.07e−1) −	3.1807e−1 (2.30e−1) −	2.4542e−1 (1.79e−1) −	**1.3786e−2 (2.32e−2)**
		200	1.9815e−1 (3.45e−2) −	NaN (NaN)	4.5727e−1 (1.11e−1) −	4.7499e−1 (3.26e−1) −	5.0945e−1 (2.49e−1) −	5.3756e−1 (2.96e−1) −	**9.9506e−3 (1.33e−2)**
		300	2.0044e−1 (3.52e−2) −	NaN (NaN)	5.2168e−1 (1.75e−1) −	6.6649e−1 (2.49e−1) −	6.1995e−1 (2.07e−1) −	7.2909e−1 (1.98e−1) −	**1.9375e−2 (2.09e−2)**
DAS-CMOP4	2	100	NaN (NaN)	NaN (NaN)	4.3396e−1 (2.25e−1) −	1.2013e+0 (0.00e+0) ≈	1.1208e+0 (1.48e−1) −	NaN (NaN)	**4.3059e−2 (1.44e−1)**
		200	NaN (NaN)	NaN (NaN)	**1.7993e−1 (2.25e−1)** ≈	NaN (NaN)	1.1364e+0 (0.00e+0) ≈	NaN (NaN)	1.8750e−1 (3.65e−1)
		300	NaN (NaN)	NaN (NaN)	3.3021e−1 (3.85e−1) ≈	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.1445e−1 (1.64e−1)**
DAS-CMOP5	2	100	NaN (NaN)	NaN (NaN)	2.4948e−1 (3.13e−1) ≈	7.4224e−1 (0.00e+0) ≈	8.6928e−1 (1.21e−1) −	NaN (NaN)	**4.3607e−2 (7.40e−2)**
		200	NaN (NaN)	NaN (NaN)	**2.8280e−2 (4.31e−2)** ≈	NaN (NaN)	1.0557e+0 (1.03e−1) −	NaN (NaN)	5.4225e−1 (4.50e−1)
		300	NaN (NaN)	NaN (NaN)	5.9801e−1 (2.14e−2) ≈	NaN (NaN)	NaN (NaN)	NaN (NaN)	**4.4288e−3 (0.00e+0)**
DAS-CMOP6	2	100	8.7851e−1 (2.12e−1) −	1.0682e+0 (9.44e−2) −	6.0363e−1 (4.67e−1) −	8.7740e−1 (2.11e−1) −	9.0632e−1 (8.70e−2) −	NaN (NaN)	**6.3999e−2 (1.18e−1)**
		200	NaN (NaN)	NaN (NaN)	**4.0164e−1 (3.24e−1)** ≈	NaN (NaN)	1.0465e+0 (0.00e+0) ≈	NaN (NaN)	7.6849e−1 (4.46e−1)
		300	NaN (NaN)	NaN (NaN)	5.7845e−1 (3.61e−1) ≈	NaN (NaN)	NaN (NaN)	NaN (NaN)	**1.3579e−2 (0.00e+0)**
DAS-CMOP7	3	100	NaN (NaN)	NaN (NaN)	1.0148e+0 (2.70e−1) −	NaN (NaN)	NaN (NaN)	NaN (NaN)	**4.6481e−2 (4.11e−2)**
		200	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**5.0909e−1 (4.66e−1)**
		300	NaN (NaN)	NaN (NaN)	7.2137e−1 (4.56e−1) ≈	NaN (NaN)	NaN (NaN)	NaN (NaN)	**4.5171e−1 (5.15e−1)**
DAS-CMOP8	3	100	NaN (NaN)	NaN (NaN)	6.3162e−1 (5.04e−1) −	NaN (NaN)	NaN (NaN)	NaN (NaN)	**4.4464e−2 (3.21e−2)**
		200	NaN (NaN)	NaN (NaN)	2.8869e−1 (3.68e−1) ≈	NaN (NaN)	NaN (NaN)	NaN (NaN)	**2.4933e−1 (2.27e−1)**
		300	NaN (NaN)	NaN (NaN)	**9.7637e−2 (1.11e−1)** ≈	NaN (NaN)	NaN (NaN)	NaN (NaN)	4.7816e−1 (4.70e−1)
DAS-CMOP9	3	100	6.8704e−1 (1.34e−1) −	NaN (NaN)	2.7186e−1 (1.83e−2) −	3.8232e−1 (8.80e−2) −	3.4729e−1 (8.39e−2) −	3.2088e−1 (4.39e−2) −	**1.8385e−1 (6.97e−2)**
		200	1.3500e+0 (0.00e+0) ≈	NaN (NaN)	4.1202e−1 (7.89e−2) −	4.6136e−1 (7.70e−2) −	3.9343e−1 (9.48e−2) −	3.6998e−1 (6.77e−2) −	**9.8198e−2 (8.89e−2)**
		300	NaN (NaN)	NaN (NaN)	4.7433e−1 (5.23e−2) −	4.8304e−1 (6.45e−2) −	4.5884e−1 (6.42e−2) −	3.7982e−1 (6.95e−2) −	**5.1979e−2 (6.57e−2)**
+/ − / ≈			0/26/1	0/27/0	0/17/10	0/25/2	0/25/2	0/27/0	

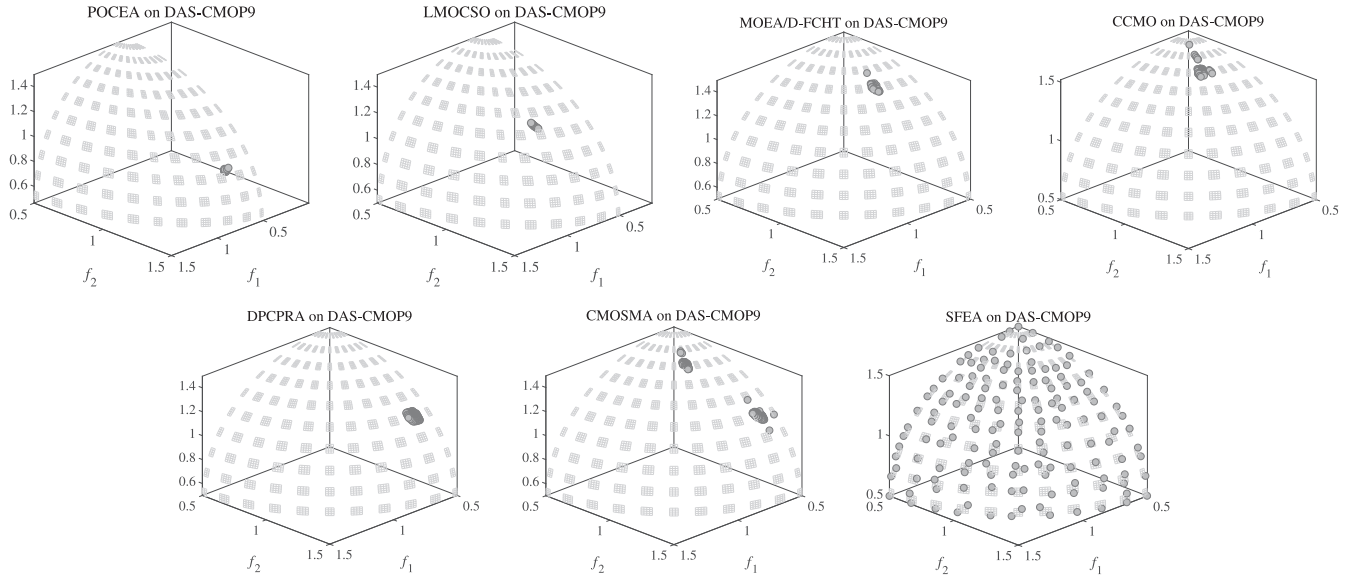


Fig. 5. Final solution set with median IGD$^+$ value among 25 runs of seven algorithms on DAS-CMOP9 with $M = 3, D = 300$.

SFEA obtained the best results in 23 out of 27 instances, except on DAS-CMOP4–6 with 200 decision variables and DAS-CMOP8 with 300 decision variables. In these four instances, MOEA/D-FCHT achieved the best performance, with SFEA slightly inferior to it. This may be because the fuzzy advantage adopted by MOEA/D-FCHT is appropriate on these few instances. However, for the remaining five peer algorithms (i.e., POCEA, LMOCSO, CCMO, DPCPRA, and CMOSMA), none of them achieved any best-performing results. For POCEA, it primarily relies on a pairwise offspring generation mechanism, lacking a dedicated constraint-handling mechanism for infeasible solutions. Additionally, it still relies on a particle updating strategy, necessitating a substantial number of evaluations to converge to the CPF. Regarding LMOCSO, it is designed for unconstrained large-scale MOPs and employs a basic CHT (i.e., CDP). The complex infeasible regions of CMOPs leads to a significant degradation in the algorithm's performance. As for CCMO, it performs unsatisfactorily on CMOPs with high-dimensional decision variables and CMOPs where the CPF is considerably distant from the UPF. For DPCPRA, it employs a multi-objective-based approach to explore infeasible regions, which does not fully utilize infeasible solutions. Regarding CMOSMA, the self-organizing mapping process may require substantial computational resources, particularly during the early stages, which could impact the algorithm's efficiency. Additionally, the final populations with the median IGD$^+$ obtained by the seven algorithms on the DAS-CMOP9 with three objectives and 500 decision variables are displayed in Fig. 5. Among seven algorithms, only SFEA exhibits good convergence and diversity.

Overall, in terms of performance on DAS-CMOPs, the proposed SFEA stands out as the best algorithm among the seven algorithms. This could be attributed to the strategy of adaptively adjusting the evolutionary fuzzy stage rationally allocates computing resources and overcomes the feasibility, convergence, and diversity difficulties caused by instances on DAS-CMOPs.

5.3. DOC

Table 5 summarizes the performance results of seven algorithms on the DOC test suite with respect to the IGD$^+$ indicator. The proposed

Table 5
IGD$^+$ results achieved by POCEA, LMOCSO, MOEA/D-FCHT, CCMO, DPCPRA, CMOSMA, and SFEA on DOC test suite.

Problem	*M*	*D*	POCEA	LMOCSO	MOEA/D-FCHT	CCMO	DPCPRA	CMOSMA	SFEA
DOC1	2	6	3.6041e−1 (9.95e−2) −	NaN (NaN)	9.9945e+2 (3.69e+2) −	3.9071e+0 (3.01e+0) −	3.5485e+0 (2.53e+0) −	5.7560e+0 (3.89e+0) −	**4.0099e−3 (7.39e−4)**
DOC2	2	16	**1.0954e−1 (0.00e+0)** ≈	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	1.5759e−1 (1.50e−1)
DOC3	2	10	8.5398e+2 (3.65e+2) −	NaN (NaN)	NaN (NaN)	6.9626e+2 (1.64e+2) −	5.8070e+2 (2.44e+2) −	5.9387e+2 (2.02e+2) −	**2.6508e−2 (2.67e−2)**
DOC4	2	8	5.8289e+1 (1.98e+2) −	NaN (NaN)	5.2131e+2 (5.10e+2) −	1.0289e+0 (6.47e−1) −	1.4900e+0 (2.34e+0) −	9.4112e−1 (1.11e+0) −	**3.9349e−2 (6.57e−3)**
DOC5	2	8	4.3351e+1 (6.29e+1) ≈	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	**3.7509e+1 (6.38e+1)**
DOC6	2	11	1.0704e+1 (1.62e+1) −	NaN (NaN)	NaN (NaN)	3.2795e+0 (2.92e+0) −	2.3739e+0 (1.31e+0) −	2.0999e+0 (2.15e+0) −	**2.0179e−1 (2.26e−1)**
DOC7	2	11	5.3384e+0 (2.29e+0) −	NaN (NaN)	NaN (NaN)	5.6172e+0 (2.03e+0) −	5.2773e+0 (2.28e+0) −	5.9499e+0 (1.82e+0) −	**8.3746e−1 (1.14e+0)**
DOC8	3	10	NaN (NaN)	NaN (NaN)	NaN (NaN)	9.4097e+1 (7.87e+1) −	7.3095e+1 (6.33e+1) −	7.6290e+1 (5.52e+1) −	**5.5960e+0 (2.04e+0)**
DOC9	3	11	NaN (NaN)	NaN (NaN)	NaN (NaN)	1.5750e−1 (1.31e−1) ≈	1.2994e−1 (1.20e−1) ≈	**4.7816e−2 (6.06e−2)** +	7.7512e−2 (1.29e−2)
+/ − / ≈			0/7/2	0/9/0	0/9/0	0/8/1	0/8/1	1/8/0	

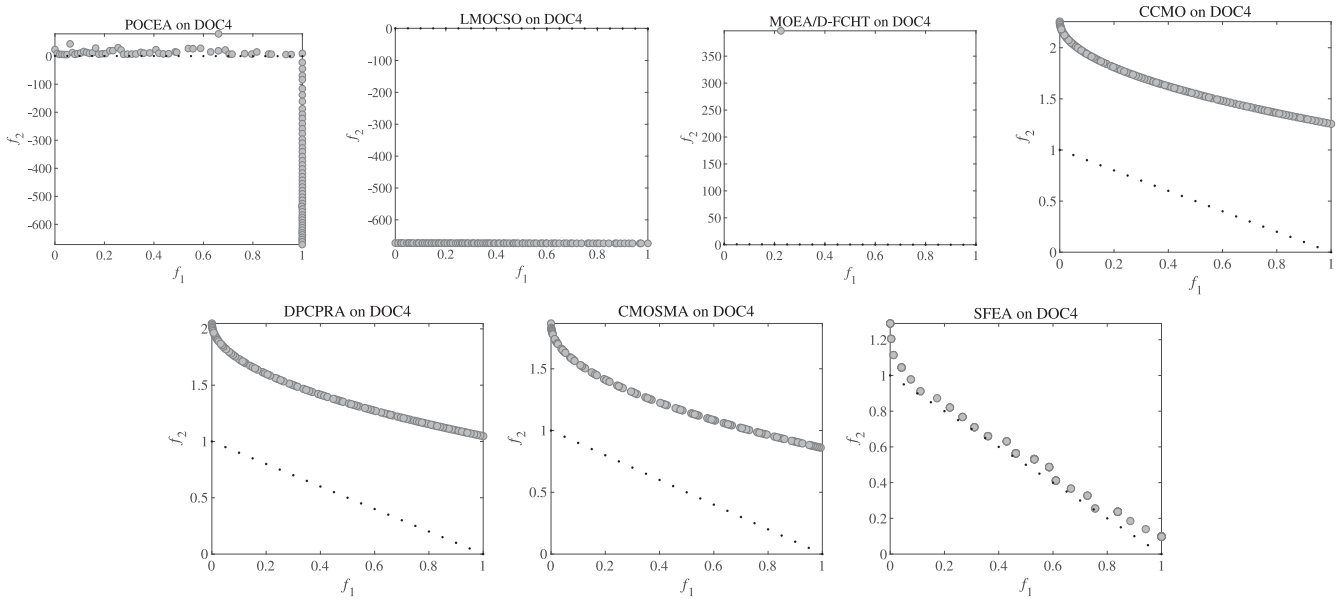


Fig. 6. Final solution set with median IGD$^+$ value among 25 runs of seven algorithms on DOC4.

SFEA is the most effective CMOEA, obtaining the best performance in 7 of the 9 test instances, except on two instances (i.e., DOC2 and DOC9).

Currently, DOC is one of the most challenging constrained test suites, featuring complex constraints within both decision space and objective space. As shown in Table 5, the proposed SFEA obtained feasible solutions on all nine test instances of DOC. In contrast, the remaining peer CMOEAs encountered instances where they failed to obtain any feasible solution in two or more problem instances. Furthermore, Fig. 6 shows the final populations with the median IGD$^+$ obtained by the seven algorithms on the DOC problem. Compared to other peer algorithms, the proposed SFEA effectively covers the CPF. The outstanding performance of SFEA is mainly attributed to the fuzzification of constraint violators by the fuzzy operator. Particularly, during the early stages of evolution, a higher degree of fuzziness is applied, leveraging promising infeasible solutions as bridges to approach the regions where optimal solutions are located. Additionally, two populations approximate the true PF from both feasible and infeasible directions, effectively maintaining population diversity.

Overall, the proposed SFEA demonstrates very promising performance in solving artificial test CMOPs with complex equality and/or inequality constraints.

5.4. Real-world CMOPs

Table 6 presents the results of seven algorithms on real-world application concerning the HV indicator. As shown, the results of the SFEA exhibit a significant superiority over all comparable CMOEAs. In terms of HV indicator, MOEA/D-FCHT obtained two best results, with LMOCSO, CCMO, DPCPRA, and CMOSMA achieved one best performance, respectively. Notably, POCEA failed to achieve any best results. From a statistical analysis perspective, SFEA significantly outperforms POCEA, LMOCSO, MOEA/D-FCHT, CCMO, DPCPRA, and CMOSMA in 15, 14, 13, 10, 9 and 9 instances. Conversely, other algorithms significantly outperform the proposed SFEA in no more than four CMOPs.

Experimental results illustrate that SFEA is not only suitable for solving CMOPs in artificial benchmarking suites, but also very effective for CMOPs in real-life applications. The proposed SFEA has the potential to serve as an alternative for solving practical CMOPs.

5.5. Parameter analysis

(1) *Analysis of parameter* κ: To analyze the impact of parameter κ on the experimental results, we conducted a parameter analysis of the proposed SFEA on the C2-DTLZ2 and DAS-CMOP1 problems. For C2-DTLZ2, its constraints only affect the objective space; whereas for DAS-CMOP1, its constraints simultaneously affect the decision and objective spaces. Specifically, we uniformly sampled the parameter κ in the range [0, 1]: 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, and 1.0.
The mean IGD$^+$ value of the proposed SFEA over 25 independent runs under 11 different values of κ is shown in Fig. 7. It is observed that, for C2-DTLZ2 with two objectives and 100 decision variables, the IGD$^+$ values obtained by the algorithm are very small across different κ values (i.e., on the order of 10^{-3}), especially in κ is greater than 0.2. This indicates that the proposed algorithm is not sensitive to the parameter κ on this instance. On the other hand, for DAS-CMOP1 with two objectives and 100 decision variables, when κ is small, the IGD$^+$ values are relatively large. Conversely, when κ is larger than 0.5, the IGD$^+$ values of the proposed SFEA are close to zero, with the minimum value achieved at $\kappa = 0.7$. To summarize, a value of 0.7 is suggested for κ.

Table 6
HV results achieved by POCEA, LMOCSO, MOEA/D-FCHT, CCMO, DPCPRA, CMOSMA, and SFEA on RWCMOPs.

Problem	POCEA	LMOCSO	MOEA/D-FCHT	CCMO	DPCPRA	CMOSMA	SFEA
RCM01	5.9209e−1 (4.57e−3) −	**9.9780e−1 (1.47e−3)** +	4.8767e−1 (2.33e−2) −	6.0732e−1 (4.49e−4) −	6.0725e−1 (4.52e−4) −	6.0698e−1 (5.41e−4) −	6.0793e−1 (3.29e−4)
RCM02	2.0365e−1 (1.13e−1) −	NaN (NaN)	2.4547e−2 (5.99e−2) −	3.5000e−1 (7.73e−2) −	3.8959e−1 (1.57e−2) −	3.7368e−1 (5.13e−2) −	**4.0078e−1 (8.73e−3)**
RCM03	3.2746e−1 (2.43e−1) −	3.6074e−1 (1.79e−1) −	8.7598e−1 (1.29e−2) −	8.9923e−1 (5.60e−4) ≈	8.9941e−1 (3.56e−4) ≈	8.9924e−1 (4.84e−4) ≈	**8.9952e−1 (4.24e−4)**
RCM04	5.1245e−3 (2.40e−2) −	5.4673e−1 (1.01e−1) −	8.5426e−1 (1.17e−3) −	**8.5958e−1 (1.67e−3)** +	8.5940e−1 (2.68e−3) +	8.5895e−1 (3.06e−3) +	8.5777e−1 (2.40e−3)
RCM05	4.2546e−1 (1.30e−3) −	2.6735e−1 (1.71e−2) −	3.9815e−1 (9.13e−3) −	4.3436e−1 (1.36e−3) −	4.3500e−1 (8.08e−5) −	4.3454e−1 (5.09e−4) −	**4.3947e−1 (3.66e−3)**
RCM06	2.7439e−1 (9.10e−4) −	NaN (NaN)	1.3667e−1 (5.31e−2) −	2.7700e−1 (3.01e−5) −	2.7700e−1 (3.00e−5) −	2.7700e−1 (4.09e−5) −	**2.8670e−1 (9.59e−3)**
RCM07	4.8080e−1 (1.21e−3) −	4.8247e−1 (2.69e−5) −	**4.8472e−1 (2.70e−5)** +	4.8456e−1 (4.50e−5) −	4.8463e−1 (4.06e−5) −	4.8439e−1 (4.80e−5) −	4.8467e−1 (3.38e−5)
RCM08	2.2824e−2 (1.62e−3) −	2.1731e−2 (1.44e−4) −	2.3473e−2 (4.24e−4) −	2.6165e−2 (5.86e−5) −	2.6202e−2 (4.17e−5) −	2.6247e−2 (2.09e−5) −	**2.7861e−2 (3.46e−5)**
RCM09	4.0898e−1 (5.39e−4) −	5.4612e−2 (8.63e−4) −	4.0241e−1 (1.99e−3) −	4.0941e−1 (1.66e−4) −	4.0957e−1 (1.18e−4) −	4.0954e−1 (1.58e−4) −	**4.1035e−1 (1.52e−4)**
RCM10	8.3841e−1 (3.70e−3) −	1.4687e−1 (1.94e−2) −	8.4726e−1 (1.07e−4) −	8.4123e−1 (2.28e−3) −	8.4189e−1 (2.07e−3) −	8.4151e−1 (1.68e−3) −	**8.7449e−1 (1.22e−3)**
RCM11	5.9920e−2 (4.01e−3) −	1.9156e−2 (3.76e−3) −	**1.0281e−1 (1.99e−4)** +	9.9461e−2 (4.38e−4) +	9.9527e−2 (4.37e−4) +	9.9432e−2 (5.86e−4) +	9.4967e−2 (1.30e−3)
RCM12	3.2852e−1 (1.21e−2) −	2.3848e−1 (7.54e−2) −	4.7502e−1 (3.55e−2) −	5.5452e−1 (4.13e−3) −	5.5637e−1 (1.52e−3) ≈	5.5536e−1 (2.98e−3) ≈	**5.5673e−1 (1.84e−3)**
RCM13	8.3154e−2 (4.07e−3) −	4.3528e−2 (1.76e−2) −	5.3512e−2 (2.04e−2) −	8.7830e−2 (1.01e−4) +	8.7825e−2 (1.40e−4) +	**8.7862e−2 (1.41e−4)** +	8.7583e−2 (1.37e−4)
RCM14	5.5083e−2 (3.65e−2) −	3.2441e−1 (1.03e−1) −	6.1135e−1 (1.84e−3) −	6.1674e−1 (1.21e−3) −	6.1736e−1 (8.51e−4) −	6.1723e−1 (5.70e−4) −	**8.4411e−1 (3.63e−3)**
RCM15	5.1990e−1 (1.17e−2) −	NaN (NaN)	4.3668e−1 (3.34e−2) −	5.4332e−1 (3.70e−4) ≈	**5.4362e−1 (5.80e−5)** +	5.4356e−1 (1.61e−4) +	5.4327e−1 (1.92e−4)
+/ − / ≈	0/15/0	1/14/0	2/13/0	3/10/2	4/9/2	4/9/2	

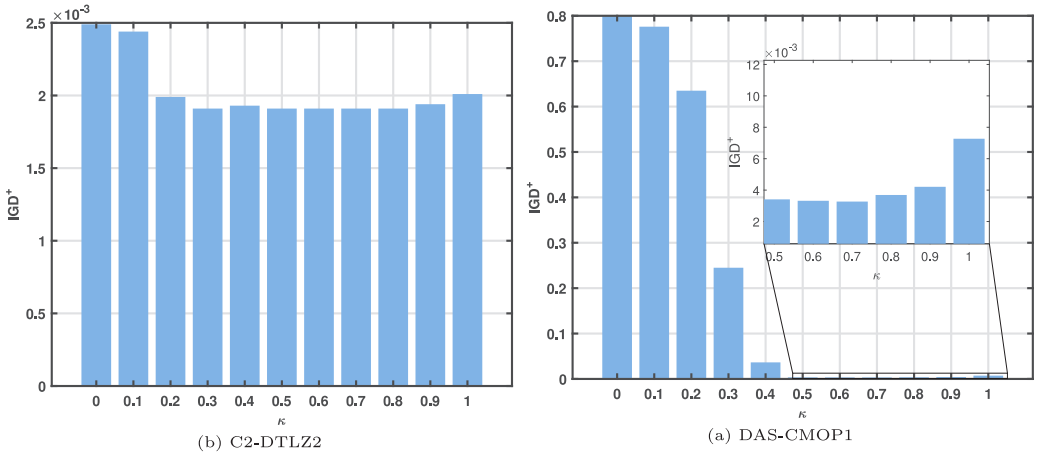


Fig. 7. Mean IGD$^+$ values of the proposed SFEA over 25 independent runs for various κ values on C2-DTLZ2 and DAS-CMOP1 with $M = 2, D = 100$.

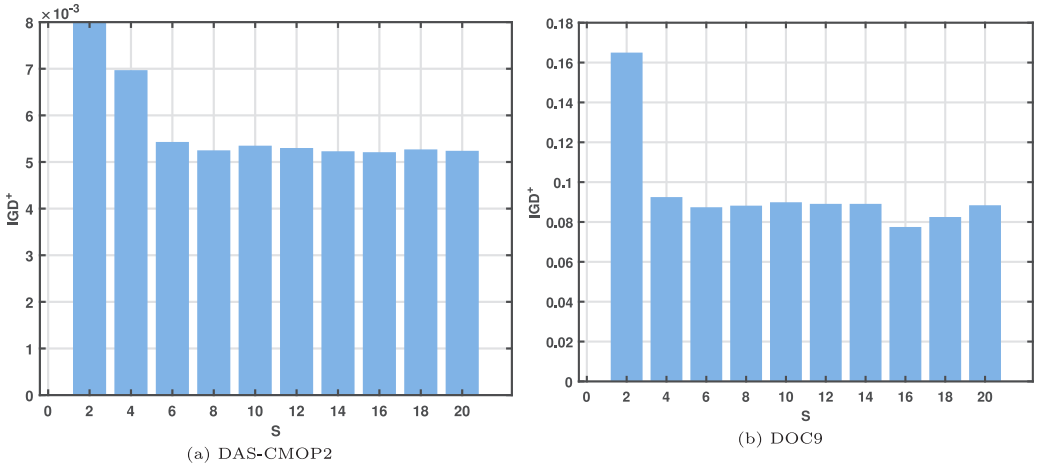


Fig. 8. Mean IGD$^+$ values of the proposed SFEA over 25 independent runs for various S values on DAS-CMOP4 with $M = 2, D = 100$, and DOC9.

(2) *Analysis of parameter S*: To investigate the impact of parameter S on experimental results, we conducted a comprehensive sensitivity analysis of the proposed SFEA on the DAS-CMOP2 and DOC9 problems. Specifically, we selected values for parameter S in the range of [2, 20] at equal intervals: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. For each selected value, the SFEA algorithm was independently run 25 times, and the average IGD$^+$ values for each case are presented in Fig. 8.

The results indicate that for the DAS-CMOP2 problem, which involves two objectives and 100 decision variables, the IGD$^+$ values are initially large as S increases from 2 to 20 but gradually decreases. Notably, when S exceeds 4, there is a marked reduction in IGD$^+$ values. This suggests that the algorithm struggles with convergence at lower S values; however, once S exceeds 4, its ability to approximate the true PF improves significantly. In the DOC9 problem, a similar trend is observed: when S is small, IGD$^+$ values are relatively high. However, once S exceeds 2, IGD$^+$ values decrease significantly and stabilize at a lower level, reaching the minimum at $S = 16$. This indicates that the SFEA demonstrates good adaptability to variations in the parameter S. Overall, the proposed SFEA exhibits strong robustness to variations in the parameter S. Additionally, the algorithm achieved

the lowest average IGD^+ value when S is set to 16, suggesting that $S = 16$ may be the optimal parameter choice.

6. Conclusion

In this work, a staged fuzzy evolutionary algorithm called SFEA for large-scale CMOPs has been proposed. In SFEA, the proposed algorithm includes two key components: fuzzy stage adjusting strategy and fuzzy operation. For the fuzzy stage adjusting strategy, aiming to balance exploration and exploitation. The early stages emphasize exploration, while the later stages shift the focus towards exploitation. Concerning the fuzzy operator, we introduced a novel variant. After normalizing the vectors, a general fuzzy operation is applied, followed by an inverse normalization. Notably, the fuzzy operator is applies to vectors of any scale dimension, making it suitable for scenarios involving large-scale optimization.

Experiments have been conducted on four benchmark CMOP test suites. Comparative analysis of the results reveals that SFEA demonstrates promising performance in addressing large-scale CMOPs. Additionally, the effectiveness of the proposed SFEA has been validated on 15 practical CMOPs. Overall, the efficacy of the novel CMOEA can be attributed to twofold: (1) The fuzzy stage adjusting strategy rationally allocates computational resources. The potential regions are explored globally in the early stages, while the optimal solutions are explored locally in the late stages. (2) Fuzzy operator includes fuzzy operations on decision variables and constraint violations. Fuzzy decision variables can help reduce the search scope and thus improve search efficiency. Fuzzy constraint violations help to utilize the information in infeasible solutions and to cross infeasible regions.

Although the proposed SFEA has shown promising results across a wide range of CMOPs, it also has limitations. (1) In adjusting the fuzzy stages, we adopted the sigmoid function. While this function is widely used in the evolutionary computation community and has been thoroughly analyzed in our parameter studies, it may not be suitable for all CMOPs. Thus, exploring alternative potential functions remains an important research direction. (2) For some test problems, such as C3-DTLZ4, particularly those that require a high level of diversity, SFEA does not perform very well. This is mainly due to the introduction of fuzzy operators, which lead to a reduction in population diversity during the early stages of evolution. Therefore, enhancing diversity preservation mechanisms in the early stages of evolution is crucial. Additionally, future research will focus on applying SFEA to address more complex real-world applications [45]. The source code for the proposed SFEA can be downloaded from the following link: https://github.com/jlzhou-coding/code.

CRediT authorship contribution statement

Jinlong Zhou: Writing – original draft, Methodology, Investigation. **Yinggui Zhang:** Visualization, Methodology, Data curation. **Fan Yu:** Writing – review & editing, Investigation. **Xu Yang:** Writing – review & editing, Conceptualization. **Ponnuthurai Nagaratnam Suganthan:** Writing – review & editing, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript.

Data availability

Data will be made available on request.

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